

Congenital Diseases of the Lower Urinary Tract



Joseph W. Bartges, DVM, PhD*, Amanda J. Callens, BS, LVT

KEYWORDS

• Congenital • Urinary bladder • Urethra • Lower urinary tract

KEY POINTS

- Congenital lower urinary tract disorders occur uncommonly in dogs and cats.
- Many congenital disorders are associated with congenital urinary incontinence.
- Treatment depends on the underlying congenital anomaly and if the anomaly is associated with clinical signs.

Genetic or acquired diseases may affect differentiation and development of the lower urinary tract. More than 400 regulatory genes seem to be involved in embryogenesis of the urinary system, which depends on a coordinated and orderly interaction of multiple embryonic tissues.^{1–3} An alteration in one or more of these genes or disruption of normal development may result in an anomaly.

URINARY BLADDER

Urinary Bladder Agenesis and Hypoplasia

Agenesis or hypoplasia of the urinary bladder results in diminished capacity for urine storage and presents as urinary incontinence.^{4–7} Embryonic maldevelopment may result in a hypoplastic bladder but is more commonly associated with ectopic ureters.⁷ Ectopic ureters may result in diminished bladder capacity, which may contribute to urinary incontinence even with surgical correction; however, bladder capacity may increase over time.^{8,9}

Pelvic Bladder

A pelvic bladder is one that has a blunt trigone, which is located in an intrapelvic location associated with a shortened urethra (**Fig. 1**).¹⁰ The role of pelvic bladder with urinary incontinence is controversial, and dogs with pelvic bladder may be continent.^{10–12} Normally, the trigone tapers and connects to the urethra in an

The authors have nothing to disclose.

Cornell University Veterinary Specialists, 880 Canal Street, Stamford, CT 06902, USA

* Corresponding author.

E-mail address: joe.bartges@cuvs.org

Vet Clin Small Anim 45 (2015) 703–719

<http://dx.doi.org/10.1016/j.cvs.2015.02.004>

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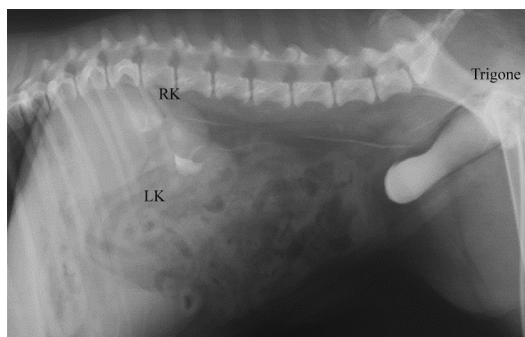


Fig. 1. Lateral abdominal radiograph of an excretory urogram in a dog with a pelvic bladder and pyelonephritis. Pyelectasia is present bilaterally. The dog had urinary incontinence and recurrent bacterial urinary tract infections. LK, left kidney; RK, right kidney; Trigone, trigone of bladder located in pelvic canal.

intra-abdominal location. The degree of distention of the bladder affects location of the trigone; therefore, the bladder should be adequately distended during contrast urethrocytography before diagnosing pelvic bladder.¹² Pelvic bladder has also been associated with urinary tract infection.¹⁰ Although some dogs with pelvic bladders are continent, other dogs with pelvic bladders have refractory urinary incontinence without any other identifiable cause.^{10,13} A diagnosis of pelvic bladder is established by contrast radiography. If pelvic bladder is associated with urinary incontinence, pharmacologic management with estrogens and/or α -adrenergic agonists should be tried before more invasive interventions. If urinary incontinence is refractory to medical therapy, urethral bulking with collagen or surgery (urethropexy or colposuspension) may be attempted.^{14–16}

Exstrophy and Herniation

Exstrophy refers to eversion of the urinary bladder, and often the intestines and external genitalia, due to absence of a portion of the ventral abdominal wall.¹⁷ This disorder is rare, being reported in an 8-month-old female English bulldog with urinary incontinence and pyelonephritis.¹⁸ Treatment involves managing associated conditions and reconstructive surgery.¹⁹

Urinary Bladder Herniation

Exteriorization of the urinary bladder through an inguinal hernia has been described in a 2-year-old cat.²⁰ Clinical signs included chronic lower urinary tract signs, and a soft mass was palpated in the region. Radiography revealed the urinary bladder to be extra-abdominal. Treatment consisted of repositioning of the urinary bladder in the abdominal cavity with an incisional cystopexy and partial closure of the enlarged inguinal canals.

Urachal Anomalies

Urachal anomalies occur commonly in dogs and cats. The urachus is a fetal connection allowing urine to pass between the developing urinary bladder and the placenta. It undergoes complete atrophy and is nonfunctional at birth. If it fails to completely atrophy, macroscopic or microscopic remnants may remain and result in persistent urachal patency or formation of urachal cysts or diverticula.^{21,22}

A persistent urachus occurs when the urachal canal remains functionally patent between the urinary bladder and the umbilicus (**Fig. 2**). It is characterized by

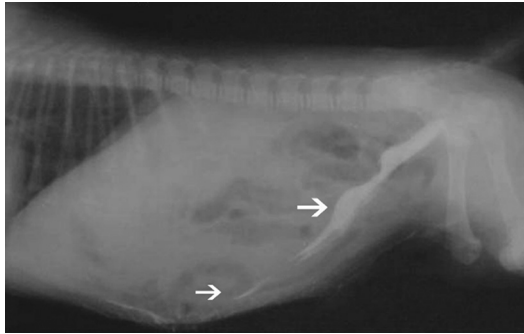


Fig. 2. Lateral contrast cystogram in an immature dog with a patent urachus (*arrows*).

inappropriate urine loss through the umbilicus.^{23–27} A patent urachus is often accompanied by omphalitis, ventral dermatitis, and urinary tract infections. Rarely, uroabdomen may occur when a persistent urachus terminates in the abdominal cavity.²⁸ A urachal cyst may develop if the urachal epithelium in an isolated segment of a persistent urachus continues to secrete fluid (**Fig. 3**).^{21,29}

A vesicourachal diverticulum occurs when a portion of the urachus located at the bladder vertex fails to close, resulting in a blind diverticulum that protrudes from the bladder apex (**Fig. 4**). Vesicourachal diverticula may be microscopic or macroscopic. Microscopic vesicourachal diverticula are microscopic lumens lined by transitional epithelium that may persist at the bladder vertex from the level of the submucosa to the subserosa.²¹ Approximately 40% of bladders from 80 cats had microscopic

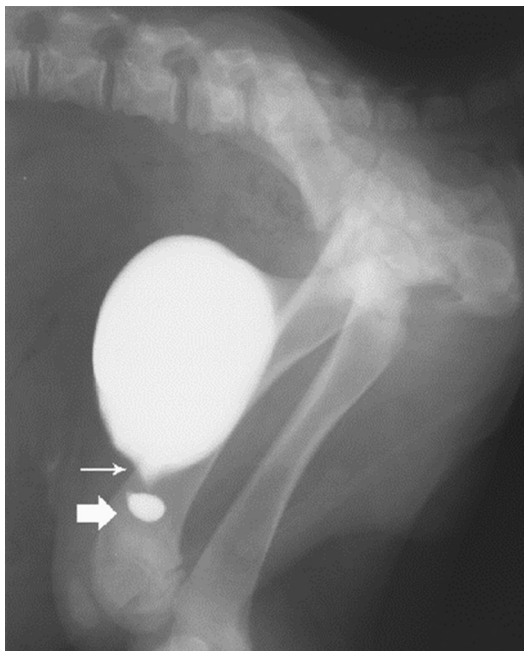


Fig. 3. Lateral contrast cystogram demonstrating a urachal cyst (*larger arrow*) and urachal diverticulum (*smaller arrow*).

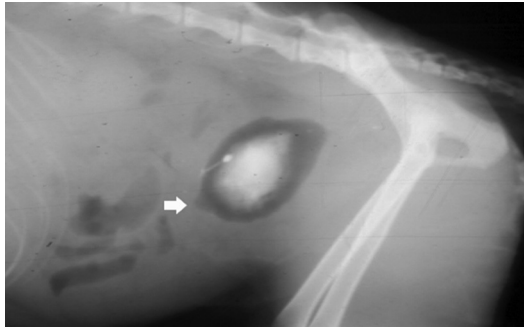


Fig. 4. Lateral double-contrast cystogram showing a urachal diverticulum (arrow) in a 3-year-old castrated male domestic shorthair cat with idiopathic cystitis.

diverticula in one study.³⁰ Usually microscopic diverticula are insignificant clinically; however, they may become macroscopic when acquired bladder and/or urethral diseases develop (eg, urinary tract infections, urolithiasis, or idiopathic cystitis).^{31–33} A microscopic diverticulum may open because of inflammation and/or increased intraluminal pressure from an acquired disease. Many of these macroscopic diverticula disappear within 2 to 3 weeks following treatment of the acquired disease and resolution of clinical signs.^{21,32} Congenital macroscopic vesicourachal diverticula are thought to be caused by urine outflow obstruction and develop before or shortly after birth. These diverticula increase the risk of bacterial urinary tract infection and associated clinical signs of lower urinary tract disease.³⁴

Vesicourachal diverticula may be visualized by contrast urethrocytography, ultrasonography, or cystoscopy. Treatment of vesicourachal diverticula depends on their size and association with clinical disease. Many macroscopic diverticula associated with active lower urinary tract disease regress with successful treatment of the acquired disease; therefore, documentation of regression of diverticulum following treatment is important.^{31,32} If the diverticulum does not resolve and recurrent disease occurs, then diverticulectomy may be warranted.^{33,35}

Urinary Bladder Duplication

Complete and partial urinary bladder duplication with and without concurrent urethral duplication has been reported in dogs.³⁶ Urinary bladder duplication may result from any alteration of normal development of the cloaca or its subdivision into the urogenital sinus and rectum.³⁷ Clinical signs develop early in life and include lower urinary tract signs, incontinence, and distended abdomen. Diagnosis is made by physical examination and imaging studies. Treatment involves surgical correction; however, success depends on degree of malformation and presence of additional congenital anomalies.

Colovesical Fistula and Uterine-Bladder Communication

Rarely, the urinary bladder may communicate with the colon or uterine horn.^{21,38,39} Clinical signs included urinary incontinence, lower urinary tract signs, and urinary tract infection. Treatment is limited to surgical correction.

Primary Urinary Bladder Neoplasia

Urinary bladder neoplasia of immature dogs and cats is rare.⁴⁰ Botryoid rhabdomyosarcoma has been observed to occur in large breed dogs younger than 18 months but can occur in other dogs.^{41–50} Botryoid rhabdomyosarcomas are embryonic

mesenchymal tumors arising from pluripotent stem cells originating from primitive urogenital ridge remnants. They are infiltrating tumors arising from the trigone and projecting into the bladder lumen as botryoid (resembling a cluster of grapes) masses.⁴⁴ Clinical signs are consistent with lower urinary tract disease, and the tumor may result in urinary outflow obstruction and hypertrophic osteoarthropathy.^{44,45} Treatment includes surgery with or without additional chemotherapy (doxorubicin, cyclophosphamide, and vincristine sulfate).^{51,52} Metastases to local tissues (lymph nodes, mesentery, omentum, prostate) and distant organs (lung, liver, kidney, spleen) have been described with this tumor.^{42,48,50}

Urocystolithiasis

Urocystolithiasis has been described in immature dogs and cats. Most commonly, struvite secondary to infection with a urease-producing microbial organism and urate secondary to congenital hepatic disease occur (**Figs. 5 and 6**).^{53–55} Although certain breeds of dogs are prone to formation of metabolic uroliths, for example, Dalmatians and urate and English bulldogs and urate or cystine, uroliths do not typically form in immature animals despite the metabolic abnormality relative to other breeds.

URETHRA

Urethral Aplasia and Hypoplasia

Urethral aplasia is a rare congenital anomaly characterized by complete absence of a patent urethra. Incontinence occurs associated with ectopic location of the ureters.^{25,56} Urethral hypoplasia described in immature female cats is associated with juvenile-onset urinary incontinence.^{5,57,58} Diagnosis is based on clinical signs and imaging studies. Radiographic features include urethral shortening and vaginal aplasia. Urethral hypoplasia may be associated with other congenital anomalies and bacterial urinary tract infection. Some affected animals respond to sympathomimetic treatment,⁵⁸ whereas in others, surgical reconstruction of the bladder neck may improve or resolve clinical signs.⁵⁷

Ectopic Ureters

Normally, the ureter enters the dorsolateral caudal surface of the bladder and empties into the trigone after a short intramural course. Ectopic ureter results from termination of one or both ureters at a site other than the trigone as a result of an embryologic abnormality of the ureteral bud of the mesonephric duct. The degree of deviation of the ureteral bud from the normal position determines the location of the ectopic opening.

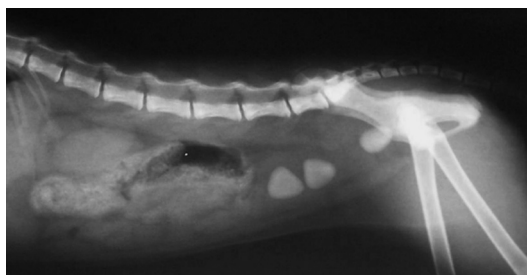


Fig. 5. Lateral abdominal radiograph of a kitten showing 3 radiopaque uroliths. The uroliths were composed of struvite that formed secondary to an infection with a urease-producing microbial organism (*Staphylococcus* spp).

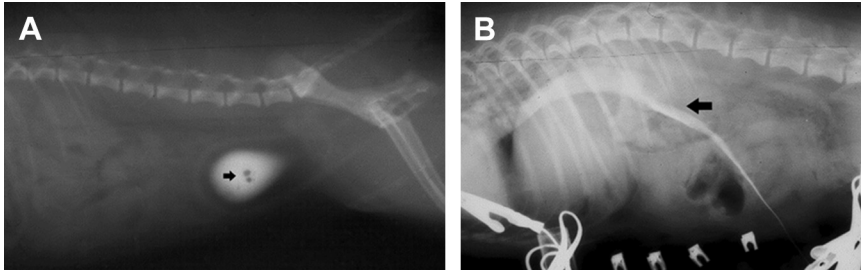


Fig. 6. (A) Lateral double-contrast cystogram of a 1-year-old castrated male Pomeranian showing urocystoliths (arrow) composed of ammonium urate that formed secondary to a congenital extrahepatic portovascular anomaly. (B) Lateral contrast portography of the dog described in (A) showing a single extrahepatic portovascular anomaly (arrow).

Ectopic ureters may be unilateral or bilateral and intramural or extramural. An extramural ectopic ureter bypasses insertion at the trigone and inserts distally in the urethra, vagina or vestibule in females, or ductus deferens in males. An intramural ectopic ureter inserts at the trigone but tunnels in the urethral wall to open distally. Variations of the intramural ectopic ureter include ureteral troughs, double ureteral openings, multiple fenestrated openings, and 2 intramural ureters opening in a single orifice.⁵⁹ Ectopic ureters may be associated with other congenital defects of the urogenital tract, including agenesis, hypoplasia, or irregular shape of the kidneys; hydroureter; ureterocele; urachal remnants, pelvic bladder; vulvovaginal strictures; and persistent hymen.^{60,61}

Young female dogs (median age of 10 months) are most commonly diagnosed with ectopic ureter; bilateral intramural ectopic ureters occur most commonly.^{6,61} Males with ectopic ureter present later, with a median age of 24 months,⁶ or they may be undiagnosed because they remain continent presumably because of the urethral length and external urethral sphincter. Breeds reported to be at greater risk for ectopic ureter include the Siberian husky, Labrador retriever, golden retriever, Newfoundland, English bulldog, West Highland white terrier, fox terrier, Skye terrier, and miniature and toy poodles.⁶¹ A genetic basis may exist but is unproven in most cases. Cats are rarely diagnosed with ectopic ureter.

Intermittent or continuous urinary incontinence since birth or weaning is the most frequently reported clinical sign in patients with ectopic ureter; however, most dogs void normally. Physical examination findings are often unremarkable, with the exception of moist or urine-stained hair in the perivulvar or preputial region. Urine scalding may cause secondary dermatitis, and owners may report frequent licking of the vulva or preputial area. Some dogs have vulvovaginitis, a vulvovaginal stricture, or a persistent hymen that can be detected digitally or with vaginoscopy. Dogs often have a history of bacterial urinary tract infections, and bacterial urinary tract infections have been reported to occur in 64% of patients.⁵⁹ Some animals may respond partially or completely to pharmacologic management of urethral sphincter mechanism incompetence.

Survey radiography should be performed to assess size, shape, and location of kidneys and bladder. Excretory urography combined with pneumocystography (Fig. 7),⁶¹ abdominal ultrasonography,⁶² contrast urethrocystography with vesicoureteral reflux, fluoroscopy, or contrast-enhanced computerized tomography may be used to diagnose ectopic ureter; however, urethrocystoscopy (Fig. 8) is more reliable especially when combined with other imaging modalities.⁶¹ Dilatation of the ectopic ureter is often but not always present.

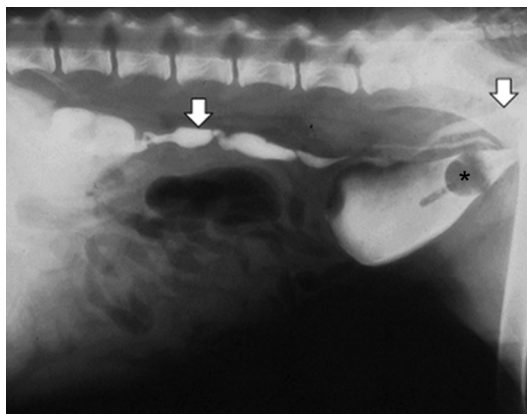


Fig. 7. Excretory urogram demonstrating a dilated extramural ectopic ureter (arrows) in a 10-month old intact female Setter. Air in the balloon of the Foley catheter (asterisk) is visible at the trigone of the urinary bladder.

Although medical management may help control urinary incontinence, surgical correction or laser ablation is the preferred treatment. An extramural ectopic ureter is ligated at the distal end of the ureter at its attachment and reimplanted into the urinary bladder between the apex and the trigone (neoureterocystostomy). The bladder wall or urine from the renal pelvis should be collected at the time of surgery and cultured because of the common occurrence of bacterial urinary tract infections.

Traditionally, intramural ectopic ureter is treated by ligating the distal submucosal ureteral segment and creating a new ureteral opening in the trigone of the urinary bladder (neoureterostomy and urethral-trigonal reconstruction); however, incontinence persists commonly (44%–67%)^{61,63,64} because the intramural segment of the ureter disrupts the functional anatomy of the internal urethral sphincter mechanism. Transurethral laser ablation may also be used to correct intramural ectopic ureter. A rigid (in females) or flexible (in males) endoscope is inserted retrograde into the urethra; diagnostic urethrocystoscopy is performed. A catheter is inserted into the lumen of the ectopic ureter to protect the urethral wall. A laser fiber, diode⁶⁵ or preferably Holmium:YAG,^{66–68} is inserted through the operating channel of the cystoscope and

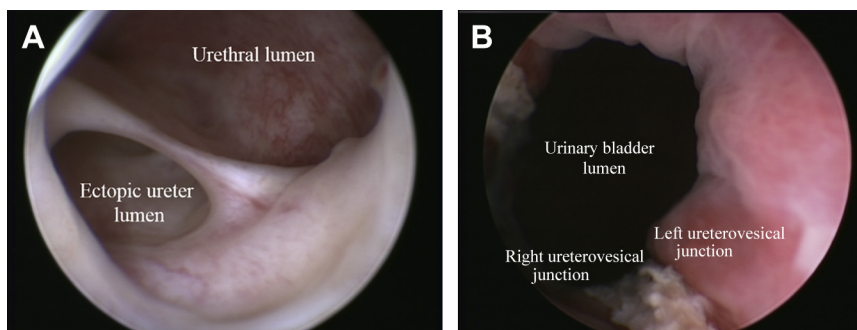


Fig. 8. (A) Cystoscopic image of a right intramural ectopic ureter in a 9-month old, intact female, Standard poodle positioned in dorsal recumbency. (B) Cystoscopic image of the right intramural ectopic ureter following laser ablation of the medial wall of the ectopic ureter to the level of the normally positioned left ureterovesical junction.

used to transect the free wall of the ectopic ureter until the opening is as close as possible to the normal anatomic location at the trigone. Incontinence may persist if the ectopic ureter is located intramurally because of disruption of the proximal urethra and internal urethral sphincter mechanism.⁶⁹ It seems that dogs are more likely to be continent with this procedure than with surgical correction. In patients in which the kidney cannot be spared because of hydronephrosis or pyelonephritis, nephroureterectomy can be performed.

Epispadias and Hypospadias

Epispadias refers to congenital defects in the dorsal aspect of the distal urethra, and hypospadias refers to anomalous ventral malposition of the urethral meatus.⁷⁰ Epispadias has been associated with exstrophy of the urinary bladder in an 8-month-old female English bulldog.¹⁸

Hypospadias occurs more commonly and usually in male dogs; Boston terriers and dalmatians have been described as having an increased risk (Fig. 9).^{70,71} It has been described in a Himalayan cat.⁷² It has been described in an East Greenland male sledge dog associated with potential in utero exposure to organochlorine, which was estimated to be 320 $\mu\text{g}/\text{d}$, corresponding to 128 pg toxic equivalents (TEQ)/kg/d. This value is 32- to 128-fold more than the World Health Organization guidelines and threshold levels for teratogen and reproductive effects.⁷³

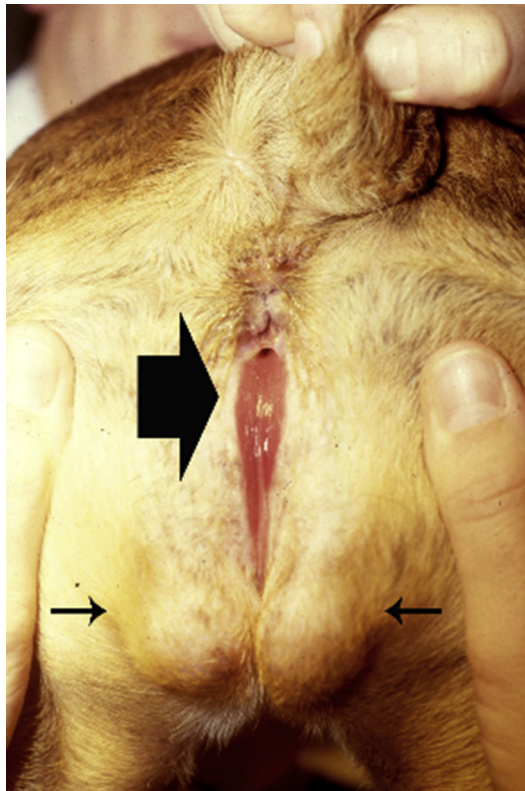


Fig. 9. Perineal hypospadias (*larger arrow*) in a 6-month-old male English bulldog. Smaller arrows designate position of the testicles.

In affected male dogs, an abnormal ventral urethral meatus may be located anywhere along the shaft of the penis, scrotum, or perineum. It is usually associated with malformation of the prepuce and penis.^{74,75} Hypospadias has been described in female dogs in association with concurrent disorders of intersexuality. Embryonically, hypospadias results from incomplete fusion of the urogenital fold. Affected dogs present at various ages and may be asymptomatic or have clinical signs of urinary incontinence, periurethral dermatitis, or bacterial urinary tract infection.^{74,76} Diagnosis is often based on physical examination. The presence of an os penis in male dogs precludes surgical reconstruction in most cases. Scrotal or perineal urethrostomy combined with castration and removal of vestigial prepuce and penile tissues may be of cosmetic value. Shortening of the penis, amputation, and urethral reconstruction have also been described.^{77–80} One technique described creating an advancement flap from the dorsal mucosa of the incompletely formed prepuce and suturing it circumferentially to construct a longer distal preputial mucosa. V- to Y-plasty of the ventral abdominal skin was used to create the preputial skin overlying the mucosal flap. Urethrostomy and partial penile amputation were also performed. This technique may be considered for glandular or penile hypospadias or after resection of the ventral aspect of the distal prepuce when inadequate tissue is present for a simple 2-layer closure of the preputial mucosa and skin.⁸¹

Urethrogenital Malformations

Urethrogenital malformations associated with diseases of intersexuality, especially pseudohermaphroditism, are often associated with urinary incontinence.⁸² Pseudohermaphrodites have gonads of one sex and external genitalia resembling those of the opposite sex.⁸³ It occurs in both sexes as a result of simultaneous development of Mullerian duct derivatives (oviduct, uterus, and portions of the vagina) and masculinization of the urogenital sinus. The phenotypic appearance of the animal depends on the degree of masculinization of the urogenital sinus. Incontinence develops early in life and may be accompanied by signs of lower urinary tract disease due to a bacterial urinary tract infection. Urinary incontinence likely results from retention of urine in anomalous communications between the urethra and the genital tract and subsequent passive leakage.⁸⁴ Diagnosis is based on clinical signs and imaging studies. Urinary incontinence may resolve with surgical correction.⁸²

Prostatic urethral diverticulae have been observed in male dogs (**Fig. 10**).⁸⁵ It has been associated with a shortened and wide intrapelvic urethra, widened urinary bladder neck, and ureteral anomalies. This condition seems to be similar to a condition described in an infant with congenital obstructive posterior urethral membrane whereby a membrane extended proximally from the verumontanum toward the bladder neck. Although this often results in chronic renal disease, a milder degree of obstruction and protective pressure pop-off mechanisms have been reported resulting in dilation of the prostatic utricle.⁸⁶

Urethral Duplication

Urethral duplication is an uncommon congenital anomaly described only in immature dogs.^{87–91} Because of the close association between embryonic development of the urogenital and gastrointestinal systems, urethral duplication is almost always accompanied by other duplication anomalies. These anomalies result from abnormal sagittal midline division and subsequent parallel development of the embryonic hindgut, cloaca, rectum, or urogenital sinus.³⁷ Associated anomalies accompanying urethral duplication depends on the stage at which dysmorphogenesis occurs. Examination of affected animals may reveal anatomic abnormalities, urinary incontinence, or

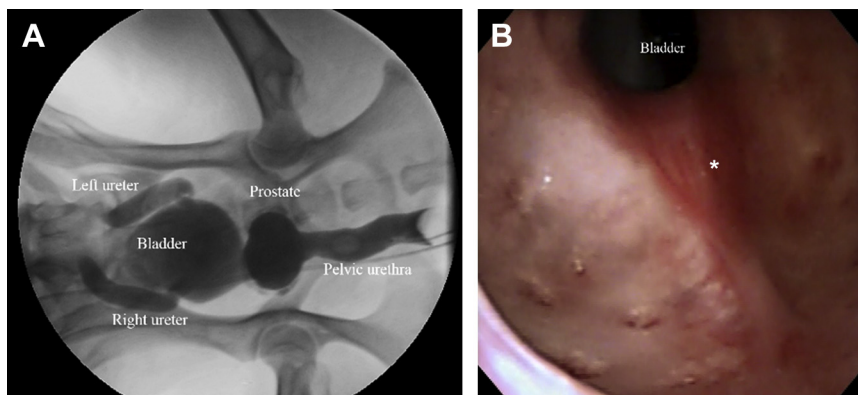


Fig. 10. Prostatic urethral diverticulæ with ureteral and urethral anomalies in a 10-month-old castrated male Labrador. (A) Ventrodorsal contrast urethrocytogram with ureteral reflux of contrast using fluoroscopy. Bilateral hydroureter is present, and the ureters enter the urinary bladder through one vesicoureteral junction. The urinary bladder neck is non-existent, and there is postprostatic urethral dilation. (B) Cystoscopic image of prostatic urethral diverticulæ. The prostatic median raphe is visible (asterisk).

clinical signs associated with secondary bacterial urinary tract infection. Diagnosis is based on physical examination, imaging studies, and exploratory surgery. Urethral duplication may in some cases be amenable to surgical extirpation of the duplicated structure; however, surgical reconstruction has rarely been attempted with extensive duplication.

Ectopic Urethra

Ectopic urethra is characterized by abnormal position of the external urethral orifice. Embryonically, urethral ectopia results from anomalous morphogenesis of the urogenital sinus, paramesonephric ducts (Mullerian ducts), or mesonephric ducts.¹ Clinical signs depend on the site of the termination of the abnormal urethra and other concurrent urogenital anomalies. Lifelong urinary incontinence was the predominant clinical feature in a 21-month-old female English bulldog with unilateral ureteral ectopia and an ectopic urethra terminating in the distal vagina.⁹² In contrast, a 2-month-old female domestic shorthair cat with ectopic urethra terminating in the ventral rectum did not have urinary incontinence but did void urine through the anus.⁹³

Urethrorectal, Urethrovaginal, and Urethroperineal Fistula and Urethral Diverticula

Fistulas connecting the urethral lumen with the large bowel, vagina, and perineal region have been described in dogs and cats.^{90,94–101} Fistula may be congenital or may be acquired because of traumatic, inflammatory, or neoplastic processes.¹⁰² Male dogs seem to be affected more frequently, and English bulldogs seem to have a predilection for urethrorectal fistula.¹⁰² Clinical signs are due to abnormal passage of urine from the fistula during urination. Additional signs may include diarrhea, perineal dermatitis, and signs associated with secondary bacterial urinary tract infections. Fistulas have been associated with infection-induced struvite urolithiasis.^{95,97,99} Diagnosis is based on clinical signs and imaging studies (Fig. 11). Treatment involves surgical correction or surgical urinary diversion. In one dog with a urethrorectal fistula secondary to a prostatic abscess, conservative therapy using an indwelling urinary catheter, low-residue diet, and antibiotics was successful.⁹⁷ Micropenis and midline

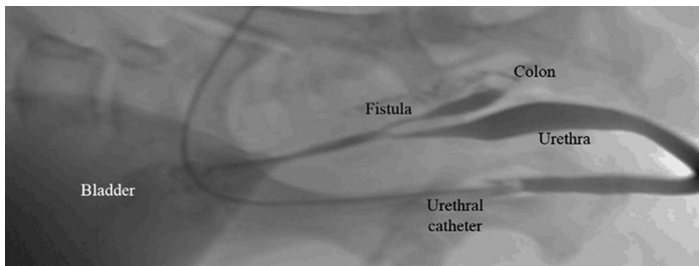


Fig. 11. Contrast urethrogram using fluoroscopy demonstrating a urethral-colonic fistula in a 1-year-old intact male English bulldog.

vestibuloperineal fistula were described in 2 dogs with urinary incontinence considered to be intersexes having 78XX karyotype.¹⁰³

Urethral Prolapse

Prolapse of the urethral mucosal lining through the external urethral orifice occurs primarily in male dogs younger than 5 years (**Fig. 12**)¹⁰⁴; English bulldogs and Boston terriers seem to have a predilection.^{105–107} Urethral prolapse may not be associated with clinical signs, or owners may only notice a red to purple mass at the tip of the penis during urination; however, it may be associated with dripping of blood, licking of the prepuce or penis, or signs of lower urinary tract disease. Diagnosis is based on physical examination. If urethral prolapse is associated with no to minimal clinical signs, treatment may not be necessary.¹⁰⁸ If associated with clinical signs, manual reduction if small or surgical correction if large may be done.^{105,107}

Urethral Stricture and Hypoplasia

Presumed congenital urethral strictures and hypoplasia have been described in young dogs and cats.^{57,109} Clinical signs relate to partial or complete urethral obstruction with stricture or urinary incontinence with hypoplasia. Systemic signs, bladder



Fig. 12. Urethral prolapse in a 2-year-old intact male English bulldog.

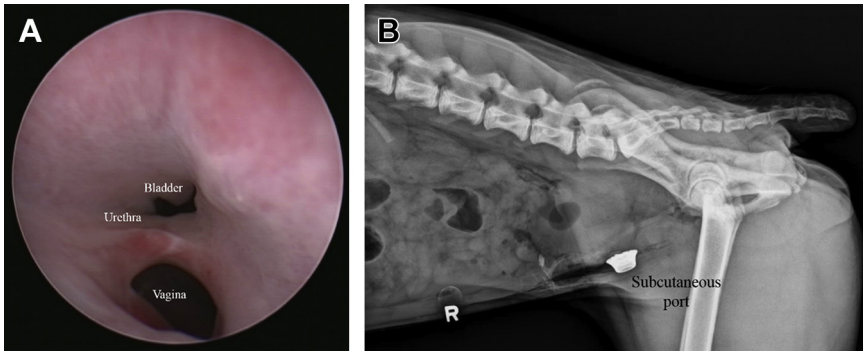


Fig. 13. Urethral hypoplasia in a 9-month-old spayed female rottweiler presented for urinary incontinence. (A) Cystoscopic image in dorsal recumbency of the urethral orifice and hypoplastic urethra with the lumen of the urinary bladder visible. The urethra measured approximately 2 cm long. (B) Immediate postoperative lateral survey abdominal radiograph demonstrating the placement of a hydraulic urethral occluder attached by tubing to the metallic subcutaneous port.

distention or rupture, overflow incontinence, secondary bacterial urinary tract infection, and hydronephrosis occur secondary to the urinary outflow obstruction or urinary incontinence with urethral hypoplasia. Urinary incontinence and bilateral hydronephrosis were observed in an 8-month-old male German shepherd with congenital midurethral stricture.¹⁰⁹ Treatment involves surgery. If the stricture occurs in the extrapelvic urethra, urethrostomy may be performed; if it occurs in the intrapelvic or intra-abdominal urethra, then urethral resection and anastomosis or prepubic urethrostomy may be indicated.^{110,111} Dilation of the urethral stricture may also be attempted with balloon or bougienage catheters. Urinary incontinence secondary to urethral hypoplasia may be treated surgically or with a hydraulic urethral occluder with variable success (Fig. 13).⁵⁷

Congenital Urinary Incontinence

Many congenital disorders are associated with congenital urinary incontinence. In addition to those described in the preceding sections, spinal dysraphism is associated with urinary and fecal incontinence. Spinal dysraphism refers to cleftlike malformations of the spine and spinal cord resulting from incomplete closure of the neural tube. Congenital urethral sphincter mechanism incompetency has been described in dogs and cats associated with urethral hypoplasia⁵⁷ or due to neurogenic causes associated with pseudohermaphroditism⁸⁴; however, ectopic ureter is most commonly associated with juvenile urinary incontinence.^{61,66,112}

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